

IMO 2026

Complete Solutions

GPT-5.6 Sol (xhigh reasoning)

Independently graded: **42/42**

Autonomous run, no human intervention.

Solutions as written by the model; all six problems.

github.com/deedy/imo-2026

IMO 2026 — Problem 1

Solution by GPT-5.6 Sol (xhigh reasoning)

2026-07-18

Source: `results/gpt-5.6-sol-xhigh/problem-01/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\text{lcm}(m, n)}{\gcd(m, n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Solution document

Status

solved

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\text{lcm}(m, n)}{\gcd(m, n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Approaches tried

- **Lexicographic descent (successful):** track the product of all board entries and the number of entries greater than 1. A move with nontrivial gcd decreases the product, while a move with gcd 1 preserves the product and decreases the number of nonunit entries. Details are in [approaches/lexicographic-descent.md](#).
- **Prime-adic exponent invariant (successful):** for each prime p , the selected exponent pair (a, b) becomes $(\min(a, b), |a - b|)$; the gcd of all p -adic exponents is consequently invariant.

Details are in [approaches/prime-adic-invariant.md](#), with the invariant isolated in [lemmas/valuation-gcd-invariant.md](#).

- **Computational sanity check:** random plays on 5000 small initial boards (four plays per board) all terminated at the value predicted by the valuation formula; see [code/random_valerify.py](#). This check is not used in the proof.

Current best

The problem is solved. If the initial integers are $x_1, \dots, x_{2026}$, then every play terminates with exactly one nonunit, and its value is explicitly

$$M = \prod_p p^{\gcd(v_p(x_1), \dots, v_p(x_{2026}))},$$

where the gcd of an all-zero list is 0. Only finitely many factors differ from 1.

Full proof

Let the entries at any stage be $y_1, \dots, y_{2026}$, and define

$$P = \prod_{i=1}^{2026} y_i, \quad K = \#\{i : y_i > 1\}.$$

All entries are positive integers throughout the process.

(a) Termination and the number of surviving nonunits

Consider a move involving $m, n > 1$, and put $d = \gcd(m, n)$. The product of the two new entries is

$$d \cdot \frac{\text{lcm}(m, n)}{d} = \text{lcm}(m, n) = \frac{mn}{d}.$$

All other entries are unchanged, so the new board product is P/d .

If $d > 1$, the positive integer P strictly decreases. If $d = 1$, the selected entries are replaced by

$$1 \quad \text{and} \quad mn.$$

In this case P is unchanged, while K decreases by exactly 1, because two entries greater than 1 have been replaced by precisely one entry greater than 1.

Suppose, for contradiction, that some sequence of moves were infinite. Along it, P would be a nonincreasing sequence of positive integers, so from some point onward P would be constant. Every move from that point onward would therefore have $d = 1$. But then the nonnegative integer K would decrease by 1 at every subsequent move, which cannot happen infinitely many times. Thus every sequence of moves is finite.

A move is possible exactly when $K \geq 2$. Hence, when Confucius can no longer move, $K \leq 1$. On the other hand, a move can never leave no entry greater than 1: if $d > 1$, its first output d is greater than 1, while if $d = 1$, its second output mn is greater than 1. Since initially $K = 2026 > 0$, we therefore have $K \geq 1$ at every stage. Consequently the terminal board has exactly one entry $M > 1$.

(b) Independence of the choices

Fix a prime p . For any positive integer z , let $v_p(z)$ denote the exponent of p in its prime factorization. We claim that

$$G_p = \gcd(v_p(y_1), \dots, v_p(y_{2026})) \quad (1)$$

is invariant under every move. Here, as usual, $\gcd(0, r) = r$, and an all-zero gcd is defined to be 0.

Indeed, let

$$a = v_p(m), \quad b = v_p(n)$$

for the two selected entries. Their two replacements have p -adic exponents

$$v_p(\gcd(m, n)) = \min(a, b)$$

and

$$v_p\left(\frac{\text{lcm}(m, n)}{\gcd(m, n)}\right) = \max(a, b) - \min(a, b) = |a - b|.$$

If, without loss of generality, $a \leq b$, then

$$\gcd(\min(a, b), |a - b|) = \gcd(a, b - a) = \gcd(a, b). \quad (2)$$

The last equality holds because an integer divides both a and $b - a$ if and only if it divides both a and b . Thus replacing a, b by $\min(a, b), |a - b|$ does not change their gcd. Taking the gcd with all the unchanged exponents proves that (1) is invariant.

Now denote the initial entries by $x_1, \dots, x_{2026}$ and set

$$g_p = \gcd(v_p(x_1), \dots, v_p(x_{2026})).$$

By part (a), the terminal entries are M and 2025 copies of 1, in some order. Applying the invariant (1) at the initial and terminal boards gives

$$g_p = \gcd(v_p(M), 0, \dots, 0) = v_p(M) \quad (3)$$

for every prime p . It follows from unique prime factorization that necessarily

$$M = \prod_p p^{g_p} = \prod_p p^{\gcd(v_p(x_1), \dots, v_p(x_{2026}))}.$$

If a prime divides none of the initial entries, its exponent in this product is 0, so the product has only finitely many nontrivial factors. The displayed value is determined solely by the initial board. Therefore the terminal value M is independent of all choices made by Confucius. $\square$

IMO 2026 — Problem 2

Solution by GPT-5.6 Sol (xhigh reasoning)

2026-07-19

Source: `results/gpt-5.6-sol-xhigh/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model. Multi-round run: earlier-round audit logs in `scratch/`.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC , respectively. Let points K and L be chosen inside triangles BMC and BNC , respectively, such that K lies inside the angle LBA , L lies inside the angle ACK , and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL . Prove that $OM = ON$.

Solution document

Status

solved

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC , respectively. Let points K and L be chosen inside triangles BMC and BNC , respectively, such that K lies inside the angle LBA , L lies inside the angle ACK , and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL . Prove that $OM = ON$.

Approaches tried

- **Angle coordinates:** Put $x = \angle KBA = \angle ACL$, $y = \angle LBK = \angle LNC$, and $z = \angle LCK = \angle BMK$. The two small triangles BMK and CNL give explicit coordinates for K and the relation between L and N .
- **Raw Cartesian and complex elimination:** Both verify the theorem, but unorganized expansion produces expressions too long for a rigorous olympiad write-up.
- **Tangent/homogeneous vector factorization:** This gives a compact exact relation between the desired circle determinant and the remaining line-incidence equation.
- **Complex unit-exponential factorization:** This independently gives the same relation. Its one-line “substitute and simplify” proof was too terse.

- **Invariant reorganization (current):** Instead of expanding the coordinates of N , write $L = \lambda N$, reduce both determinants to $K \cdot L$, $[K, L]$, and $|L|^2$, and then compare the four coefficients of the ray parameter. This yields a short invariant expansion and an explicit four-row coefficient check, completing the factorization.
- **Final proof audit (current session):** The argument has been checked through the coordinate setup, ray orientations, the incidence determinant, the power determinant (including its signs), and all exceptional tangent cases. The continuity domain includes all positivity constraints and verifies $\lambda \neq 0$; the coefficient check states an exact reproducible expansion procedure and records every resulting remainder. The factorization lemma is also recorded as a standalone proved lemma, and the Markdown/LaTeX structure has been checked. No mathematical gap or unproved geometric case remains.

Current best

A complete analytic proof is given below. After normalizing $A = 0$, $B = 2$, the three given angle equalities provide explicit coordinates for K, L, N . The remaining incidence condition is $[K - 2N, e^{i(x+z)}N] = 0$. A proved determinant factorization shows that the difference of the powers of M and N with respect to (AKL) is a scalar multiple of this incidence determinant. Hence the powers are equal, which is exactly $OM = ON$.

Full proof

We identify the Euclidean plane with $\mathbb{C}$. After a similarity and, if necessary, a reflection, take

$$A = 0, \quad B = 2, \quad M = 1,$$

with C in the upper half-plane. Lower-case letters will also denote the complex coordinates of the corresponding upper-case points. Put

$$x = \angle KBA = \angle ACL, \quad y = \angle LBK = \angle LNC, \quad z = \angle LCK = \angle BMK.$$

All three numbers are positive. Moreover, the nondegenerate triangles BMK and CNL give

$$0 < x + z < \pi, \quad 0 < x + y < \pi. \quad (1)$$

In particular, the sines occurring below are nonzero.

The conditions $K \in \text{int } \angle LBA$, $\angle KBA = x$, and $\angle LBK = y$ imply that the rays BA, BK, BL occur in this order inside $\angle LBA$. Thus the oriented angle from BA to BL , toward the interior of $\angle ABC$, is $x + y$. Since the vector BA points in direction π , there is a number $t = BL > 0$ such that

$$l = 2 - te^{-i(x+y)}. \quad (2)$$

In triangle BMK , we have $BM = 1$, the angles at B, M are x, z , respectively, and the angle at K is $\pi - x - z$. The sine rule and the direction of MK therefore give

$$k = 1 + \frac{\sin x}{\sin(x+z)} e^{iz}. \quad (3)$$

Also $C = 2N$. In triangle CNL , the angles at C, N are x, y : indeed, C, N, A are collinear and $\angle NCL = \angle ACL = x$, while $\angle LNC = y$. Thus the sine rule gives

$$\frac{NL}{CN} = \frac{\sin x}{\sin(x+y)}.$$

Because L is inside triangle BNC , it lies on the same side of NC as B ; in our orientation this is the clockwise side. Hence the ray NL is obtained from NC by a clockwise turn through y , so

$$l = n + \frac{\sin x}{\sin(x+y)} e^{-iy} n = \lambda n, \quad \lambda = 1 + \frac{\sin x}{\sin(x+y)} e^{-iy}. \quad (4)$$

Finally, because L lies inside $\angle ACK$, the rays CA, CL, CK occur in this order and therefore $\angle ACK = x + z$. The points K and L , being inside triangles BMC and BNC , lie inside ABC ; hence CK is on the interior side of CA . Since $CA = -n$, rotating CA through $x + z$ toward the interior gives the direction $-e^{i(x+z)} n$. Thus $k - 2n$ is a real multiple of $e^{i(x+z)} n$ (the sign of that multiple is immaterial). If

$$[u, v] = \operatorname{Im}(\bar{u}v),$$

this is precisely

$$Q := [k - 2n, e^{i(x+z)} n] = 0. \quad (5)$$

We shall use the following algebraic identity, proved below.

Factorization lemma. Under (2)–(4),

$$\begin{aligned} D &:= [k, l](|n|^2 - 1) - [k, n - 1]|l|^2 + [l, n - 1]|k|^2 \\ &= \frac{\sin x (t \sin x + \sin y)}{\sin(x+y) \sin(x+z)} [k - 2n, e^{i(x+z)} n]. \end{aligned} \quad (6)$$

Assuming the lemma, (5) and (6) yield $D = 0$. It remains to explain why this is the desired conclusion. Let o be the coordinate of the circumcenter O . Since the circle (AKL) passes through the origin, its power function is

$$\mathcal{P}(w) = |w - o|^2 - |o|^2 = |w|^2 - 2 \operatorname{Re}(\bar{o}w).$$

The equations $\mathcal{P}(k) = \mathcal{P}(l) = 0$ are

$$2O \cdot K = |k|^2, \quad 2O \cdot L = |l|^2. \quad (7)$$

The equality $\mathcal{P}(n) = \mathcal{P}(1)$ is equivalently

$$2O \cdot (N - M) = |n|^2 - 1. \quad (8)$$

The augmented determinant of the three linear equations (7)–(8), viewed as equations in the two coordinates of $2O$, is

$$\det \begin{pmatrix} \operatorname{Re} k & \operatorname{Im} k & |k|^2 \\ \operatorname{Re} l & \operatorname{Im} l & |l|^2 \\ \operatorname{Re}(n-1) & \operatorname{Im}(n-1) & |n|^2 - 1 \end{pmatrix}.$$

Expanding it along the third column gives exactly

$$[k, l](|n|^2 - 1) - [k, n - 1]|l|^2 + [l, n - 1]|k|^2 = D.$$

The circumcenter of AKL is given, so A, K, L are noncollinear; equivalently, the first two coefficient vectors in (7) are linearly independent and determine O uniquely. Hence $D = 0$ implies (8). Therefore

$$|ON|^2 - |OM|^2 = \mathcal{P}(n) - \mathcal{P}(1) = 0,$$

and so $OM = ON$.

Verification of the factorization lemma

First suppose that $\cos y \cos z \neq 0$. We have $0 < x < \pi/2$: indeed, the ray BK is inside $\angle ABC$ and the ray CL is inside $\angle ACB$, so x is smaller than both of those angles, while the latter cannot both be at least $\pi/2$. Put

$$p = \tan x, \quad q = \tan y, \quad s = \tan z, \quad r = t \cos x \cos y. \quad (15)$$

The signs of q, s, r will not matter. By (1), $p + q$ and $p + s$ are nonzero. Dividing the coordinate formulas (2)–(4) by the appropriate products of cosines gives

$$K = \left(\frac{2p + s}{p + s}, \frac{ps}{p + s} \right), \quad L = (2 - r(1 - pq), r(p + q)), \quad (16)$$

and

$$L = \lambda N, \quad \lambda = U + iV, \quad U = \frac{2p + q}{p + q}, \quad V = -\frac{pq}{p + q}. \quad (17)$$

Set

$$W = U^2 + V^2, \quad P = K \cdot L, \quad C = [K, L].$$

Since $N = \bar{\lambda}L/W$, we obtain

$$K \cdot N = \frac{UP + VC}{W}, \quad [K, N] = \frac{UC - VP}{W}, \quad |N|^2 = \frac{|L|^2}{W}, \quad [L, N] = -\frac{V|L|^2}{W}. \quad (18)$$

Using $[K, N - 1] = [K, N] + K_y$ and $[L, N - 1] = [L, N] + L_y$ in the definition of D , and then substituting (18), yields

$$WD = C((1 - U)|L|^2 - W) + |L|^2\{V(P - |K|^2) - WK_y\} + WL_y|K|^2. \quad (19)$$

We also replace rotation through $x + z$ by the nonzero real multiple

$$T = (1 - ps) + i(p + s) = \frac{e^{i(x+z)}}{\cos x \cos z}.$$

Put $Q_T = [K - 2N, TN]$. If $T = \alpha + i\beta$, then $[X, TY] = \alpha[X, Y] + \beta X \cdot Y$. Applying this with $\alpha = 1 - ps$, $\beta = p + s$, and using (18), gives

$$WQ_T = C((1 - ps)U + (p + s)V) + P((p + s)U - (1 - ps)V) - 2(p + s)|L|^2. \quad (20)$$

For completeness, we now verify the polynomial identity obtained from (19)–(20). Abbreviate

$$d = p + q, \quad e = p + s, \quad a = 1 - pq, \quad g = (1 + p^2)(1 + q^2), \quad j = \frac{p}{d}.$$

Write $K = (u, v)$. Formula (16) gives the two useful linear relations

$$v = p(2 - u), \quad e(u - 1) = p. \quad (21)$$

Writing $P = P_0 + P_1r$ and $C = C_0 + C_1r$, direct multiplication in (16) gives

$$P_0 = 2u, \quad P_1 = -au + dv, \quad C_0 = -2v, \quad C_1 = du + av, \quad (22)$$

whereas

$$|L|^2 = 4 - 4ar + gr^2, \quad L_y = dr, \quad U = 1 + j, \quad V = -qj, \quad W = 1 + 2j + j^2(1 + q^2). \quad (23)$$

Define

$$X = C_0 + q(P_0 - |K|^2), \quad Y = C_1 + qP_1, \quad Z = -jX - Wv, \quad Z_1 = -jY. \quad (24)$$

From (19) and (22)–(24), collection by powers of r gives

$$WD = A_0 + A_1r + A_2r^2 + A_3r^3, \quad (25)$$

where

$$\begin{aligned} A_0 &= -WC_0 + 4Z, \\ A_1 &= -WC_1 - 4aZ + 4Z_1 + Wd|K|^2, \\ A_2 &= gZ - 4aZ_1, \\ A_3 &= gZ_1. \end{aligned} \quad (26)$$

Similarly, on putting

$$\begin{aligned} \alpha &= (1 - ps)U + eV = (1 - p(e - p))(1 + j) - eqj, \\ \beta &= eU - (1 - ps)V = e(1 + j) + (1 - p(e - p))qj, \end{aligned}$$

formula (20) gives

$$WQ_T = B_0 + B_1r + B_2r^2, \quad (27)$$

where

$$B_0 = C_0\alpha + P_0\beta - 8e, \quad B_1 = C_1\alpha + P_1\beta + 8ea, \quad B_2 = -2eg. \quad (28)$$

Thus

$$deWD = p(q + p(1 + q^2)r)WQ_T \quad (29)$$

is equivalent to the following four coefficient equalities:

$$\begin{aligned} deA_0 &= pqB_0, \\ deA_1 &= p\{qB_1 + p(1 + q^2)B_0\}, \\ deA_2 &= p\{qB_2 + p(1 + q^2)B_1\}, \\ deA_3 &= p^2(1 + q^2)B_2. \end{aligned} \quad (30)$$

Here is an explicit check, included to leave no algebra hidden. Let R_i be the left side minus the right side in the i -th equality in (30), numbered from 0 to 3. Substitution from (21)–(28), using first $v = p(2 - u)$, gives

$$R_0 = \frac{4pq(1 + p^2)}{d}(e(u - 1) - p)(du - 2p - q), \quad (31)$$

$$R_1 = \frac{1 + p^2}{d}(e(u - 1) - p)F_1, \quad R_2 = \frac{p(1 + p^2)(1 + q^2)}{d}(e(u - 1) - p)F_2, \quad R_3 = 0, \quad (32)$$

where

$$\begin{aligned} F_1 &= 5p^3q^2u - 10p^3q^2 + 4p^3u - 8p^3 + 5p^2q^3u - 4p^2q^3 + 4p^2qu \\ &\quad + pq^2u + 2pq^2 + q^3u, \\ F_2 &= p^3qu - 2p^3q + p^2q^2u + 4p^2 + pqu + 2pq + q^2u. \end{aligned}$$

To verify (31)–(32) without any concealed division, multiply each R_i first by d and insert

$$v = p(2 - u), \quad U = 1 + \frac{p}{d}, \quad V = -\frac{pq}{d}, \quad W = 1 + \frac{2p}{d} + \frac{p^2(1 + q^2)}{d^2},$$

together with (22)–(28); collecting the coefficients of u gives exactly the displayed products. This is ordinary distributive expansion, and the formulas (31)–(32) record the full remainders after collection. The second relation in (21) makes every one of them zero, proving (30), hence (29).

Because $W \neq 0$ and $de \neq 0$, (29) gives

$$D = \frac{p\{q + p(1 + q^2)r\}}{de} Q_T. \quad (33)$$

Now

$$Q_T = \frac{Q}{\cos x \cos z}$$

and, using (15),

$$\frac{p\{q + p(1 + q^2)r\}}{(p + q)(p + s) \cos x \cos z} = \frac{\sin x (t \sin x + \sin y)}{\sin(x + y) \sin(x + z)}. \quad (34)$$

Indeed, substitute $p = \sin x / \cos x$, $q = \sin y / \cos y$, $s = \sin z / \cos z$, and $r = t \cos x \cos y$; cancellation gives (34). Equations (33)–(34) prove (6) when $\cos y \cos z \neq 0$.

Finally, both sides of (6), with k, l, n, λ defined by (2)–(4), are continuous functions of (x, y, z, t) on the open set

$$0 < x < \frac{\pi}{2}, \quad y > 0, \quad z > 0, \quad x + y < \pi, \quad x + z < \pi, \quad t > 0.$$

(Here $0 < y < \pi - x$, so $\operatorname{Im} \lambda = -\frac{\sin x \sin y}{\sin(x+y)} < 0$; in particular $\lambda \neq 0$.) The subset on which $\cos y \cos z \neq 0$ is dense in this open set. Thus (6) also holds when $\cos y \cos z = 0$, by continuity. This completes the proof of the factorization lemma and of the theorem.

IMO 2026 — Problem 3

Solution by GPT-5.6 Sol (xhigh reasoning)

2026-07-18

Source: `results/gpt-5.6-sol-xhigh/problem-03/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n , determine the largest value c such that Liu Bang may guarantee a total length of at least c , regardless of Xiang Yu's play.

Solution document

Status

solved

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n , determine the largest value c such that Liu Bang may guarantee a total length of at least c , regardless of Xiang Yu's play.

Approaches tried

- Reduced the claiming stage to the alternating discrepancy of the sorted final lengths. If $x_1 \geq \dots \geq x_k$, the draft value for Liu Bang is $x_1 + x_3 + \dots = (1 + D)/2$, where $D = x_1 - x_2 + x_3 - \dots$.
- Obtained the preliminary upper bound $D \leq 1/(2n + 1)$ by making equal pairs of pieces; this is not sharp for $n \geq 2$.

- Tested several candidate initial partitions for small n . For $n = 2$, the failed candidate proportional to $(1, 1, 3)$ and the successful discrepancy $1/7$ suggested a powers-of-two construction.
- Recast the lower bound as a sparse weighted-multigraph problem: paired final pieces become weighted edges between their original Liu Bang intervals. A tree component forces a nonzero signed sum of powers of two.
- Proved the matching upper bound by pigeonholing the 2^{n+1} subset sums of arbitrary initial interval lengths, then realizing two close subset sums as equal pairs using a common refinement with at most n further cuts.
- Ran random checks for $1 \leq n \leq 8$ on the powers-of-two lower construction and on the subset-sum gap bound; these supported the proved formula but were not used in place of proof.

Current best

The explicit answer is $c = \frac{2^n}{2^{n+1} - 1}$. Equivalently, the optimal alternating discrepancy between the two players' totals is $1/(2^{n+1} - 1)$. Liu Bang attains it by making initial pieces proportional to $1, 2, 4, \dots, 2^n$, while Xiang Yu's matching strategy follows from two subset sums of the initial piece lengths differing by at most $1/(2^{n+1} - 1)$.

Full proof

Put

$$d = \frac{1}{2^{n+1} - 1}.$$

We first establish three facts that will be used in both directions.

1. Value of the claiming stage

Suppose that, after all cuts, the piece lengths in nonincreasing order are

$$x_1 \geq x_2 \geq \dots \geq x_k > 0,$$

and define their **alternating discrepancy** by

$$D(x_1, \dots, x_k) = x_1 - x_2 + x_3 - x_4 + \dots.$$

We claim that under optimal play the difference between Liu Bang's total and Xiang Yu's total is exactly D .

For a multiset X of lengths, let $V(X)$ be the maximum difference between the current player's eventual total and the other player's eventual total. Backward induction gives

$$V(X) = \max_{x \in X} (x - V(X \setminus \{x\})). \quad (1)$$

We prove by induction on $|X|$ that $V(X) = D(X)$, where the elements of X are put in nonincreasing order when D is evaluated. This is immediate for the empty multiset. Suppose the result holds for smaller multisets and write $X = (x_1, \dots, x_k)$. Choosing x_1 in (1) gives

$$x_1 - D(x_2, \dots, x_k) = D(x_1, \dots, x_k).$$

If instead x_j is chosen, direct cancellation shows that

$$D(X) - (x_j - D(X \setminus \{x_j\})) = \begin{cases} 2((x_1 - x_2) + \cdots + (x_{j-2} - x_{j-1})), & j \text{ odd}, \\ 2((x_1 - x_2) + \cdots + (x_{j-3} - x_{j-2}) + (x_{j-1} - x_j)), & j \text{ even}. \end{cases} \quad (2)$$

Here an empty sum is 0. Every term on the right is nonnegative, so no choice is better than x_1 . This proves the claim.

The two players' totals sum to 1. Therefore Liu Bang's value in the claiming stage is

$$\frac{1 + D}{2}. \quad (3)$$

Thus it remains to prove that the minimax value of D in the marking stage is d .

2. A pairing interpretation of D

If there are an even number of lengths, the minimum, over all partitions of the pieces into pairs, of the sum of the absolute differences within the pairs is D . If there are an odd number, the analogous minimum—pair all but one piece and add the singleton's length to the pairwise differences—is again D .

To prove this, first consider $2r$ lengths. In any pairing in which x_1 and x_2 are not together, suppose their pairs are (x_1, x_j) and (x_2, x_ℓ) . Replacing these by (x_1, x_2) and (x_j, x_ℓ) does not increase the cost: if $\ell < j$, the old cost minus the new cost is

$$(x_1 - x_j + x_2 - x_\ell) - (x_1 - x_2 + x_\ell - x_j) = 2(x_2 - x_\ell) \geq 0,$$

and if $j < \ell$ it is similarly $2(x_2 - x_j) \geq 0$. Induction therefore shows that the adjacent pairing

$$(x_1, x_2), (x_3, x_4), \dots, (x_{2r-1}, x_{2r})$$

has minimum cost, namely D . For an odd number of lengths, append an extra length 0 and apply the even case; pairing a piece with 0 is precisely the same as declaring that piece the singleton.

In particular, if the final pieces can be grouped into equal pairs together with some remaining pieces whose total length is at most d , then

$$D \leq d. \quad (4)$$

Indeed, retain the equal pairs, pair the remaining pieces arbitrarily, and leave one as a singleton if necessary. Since $|u - v| \leq u + v$, this pairing has cost at most the total length of the remaining pieces.

3. A common-refinement construction

We need the following cutting fact. Let A and B be two disjoint nonempty collections of initial intervals, with total lengths $p \geq q > 0$. Then using at most

$$|A| + |B| - 1 \quad (5)$$

cuts within these intervals, it is possible to create equal pairs, each containing one piece from an interval of A and one from an interval of B , while leaving unpaired material only in A , of total length $p - q$.

To see this, order the intervals of A . Take them in that order until a total length q has been selected, using an initial portion of the last selected interval if necessary. Suppose r intervals of A are involved. Concatenate their selected portions to make one abstract interval $[0, q]$, and concatenate the $s = |B|$ intervals to make another copy of $[0, q]$. Superimpose the two sets of division points. The common refinement has at most

$$E \leq r + s - 1$$

cells. Cut the physical intervals at these refinement points and pair the corresponding cells of the two copies.

On the B side this requires $E - s$ cuts. If the last selected A -portion is a whole interval, the A side requires $E - r$ cuts, for a total at most $r + s - 2$. If that portion is proper, its unused tail contributes one additional piece, so the total number of cuts is

$$(E - s) + (E - r + 1) \leq r + s - 1 \leq |A| + |B| - 1.$$

All material not in an equal pair lies on the A side and has total length $p - q$, proving the cutting fact.

4. Liu Bang can force $D \geq d$

Liu Bang makes exactly n marks so that the resulting $n + 1$ consecutive intervals have lengths

$$d, 2d, 4d, \dots, 2^n d. \quad (6)$$

They have total length $(2^{n+1} - 1)d = 1$.

Consider any response by Xiang Yu using at most n marks. After all the cuts, arrange the final pieces in adjacent pairs that attain the pairing cost D , as in Section 2, with one singleton if necessary. Construct a weighted multigraph whose vertices $0, 1, \dots, n$ correspond to Liu Bang's initial intervals. For each pair of final pieces of lengths u, v , draw an edge between the vertices of their initial intervals and give it weight $\min(u, v)$. If both pieces came from the same initial interval, this is a loop, whose weight is counted twice in the weighted degree. For each pair, regard the excess $|u - v|$ as residual material at the vertex containing the larger piece; a singleton is entirely residual.

There are at most $2n + 1$ final pieces, so this graph has at most n edges. If $\deg_w(i)$ is the weighted degree of vertex i and $r_i \geq 0$ is its total residual, then all material from initial interval i has been accounted for, and hence

$$2^i d = \deg_w(i) + r_i. \quad (7)$$

Moreover, by the definition of the adjacent pairing,

$$D = \sum_{i=0}^n r_i. \quad (8)$$

A multigraph on $n + 1$ vertices with at most n edges has a connected component which is a tree, where an isolated vertex is allowed. Indeed, every connected component with v vertices has at least $v - 1$ edges; if none had exactly $v - 1$, the total number of edges would be at least the total number of vertices, namely $n + 1$.

Let $P \sqcup Q$ be the bipartition of such a tree component. It has no loops, and every edge has one endpoint in each part, so

$$\sum_{i \in P} \deg_w(i) = \sum_{i \in Q} \deg_w(i).$$

Subtracting (7) over its two parts gives

$$d \left(\sum_{i \in P} 2^i - \sum_{i \in Q} 2^i \right) = \sum_{i \in P} r_i - \sum_{i \in Q} r_i. \quad (9)$$

The integer in parentheses is nonzero: in a nonempty signed sum of distinct powers of two, the largest power is greater than the sum of all smaller powers. Its absolute value is therefore at least 1. It follows from (9) that

$$d \leq \left| \sum_{i \in P} r_i - \sum_{i \in Q} r_i \right| \leq \sum_{i=0}^n r_i = D. \quad (10)$$

Thus Liu Bang's construction guarantees $D \geq d$.

5. Xiang Yu can force $D \leq d$

Consider any initial marking by Liu Bang, and let the positive initial interval lengths be $a_1, \dots, a_m$, where $m \leq n+1$.

If $m \leq n$, Xiang Yu bisects every initial interval. This takes at most n cuts and produces only equal pairs, so $D = 0 \leq d$ by Section 2.

It remains to consider $m = n+1$. Form all 2^{n+1} subset sums

$$\sum_{i \in S} a_i, \quad S \subseteq \{1, \dots, n+1\}.$$

They lie in $[0, 1]$. When put in nondecreasing order, two consecutive subset sums differ by at most

$$\frac{1}{2^{n+1} - 1} = d. \quad (11)$$

Choose distinct subsets S, T giving such a pair, and cancel their intersection. We obtain disjoint subsets

$$A = S \setminus T, \quad B = T \setminus S,$$

not both empty, such that

$$\left| \sum_{i \in A} a_i - \sum_{i \in B} a_i \right| \leq d. \quad (12)$$

If one of A, B is empty, let C be the nonempty one. By (12), the total length of the intervals indexed by C is at most d . Xiang Yu leaves those intervals alone and bisects every interval outside C . This uses

$$n+1 - |C| \leq n$$

cuts and leaves equal pairs together with pieces of total length at most d . Hence $D \leq d$ by (4).

Now suppose both A and B are nonempty. Apply the common-refinement construction of Section 3 to these two collections, putting the one with larger total length first. By (12), it creates equal pairs and leaves unpaired material of total length at most d , using at most $|A| + |B| - 1$ cuts. Bisect every interval outside $A \cup B$. The total number of cuts is at most

$$(|A| + |B| - 1) + (n + 1 - |A| - |B|) = n. \quad (13)$$

Again, all pieces except for material of total length at most d occur in equal pairs, so (4) yields $D \leq d$.

Thus Xiang Yu can always force $D \leq d$. Combined with Section 4, the marking-stage minimax discrepancy is exactly d . Finally, (3) gives

$$c = \frac{1+d}{2} = \frac{1}{2} \left(1 + \frac{1}{2^{n+1}-1} \right) = \boxed{\frac{2^n}{2^{n+1}-1}}.$$

IMO 2026 — Problem 4

Solution by GPT-5.6 Sol (xhigh reasoning)

2026-07-18

Source: `results/gpt-5.6-sol-xhigh/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

solved

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Approaches tried

- **Angle-coordinate model (successful):** If the current angles are (A, B, C) and a cut from the A -vertex splits A into x and $A - x$, the children have angles $(B, x, 180^\circ - B - x)$ and

$(C, A - x, B + x)$. Thus the two new angles at the cut point are supplementary.

- **Finite-horizon backward sets (partially useful):** Starting from triangles already containing θ , an attempted full description of successive winning sets became unwieldy. Its useful consequence was the recursive observation that every triangle containing $m\theta$, for a positive integer m , is winning.
- **Complementary cut-point angles (successful):** When $180^\circ = n\theta$, a cut can be chosen so that the two children have cut-point angles $k\theta$ and $(n - k)\theta$. An elementary integer-between-partial-sums lemma guarantees such a legal cut from every triangle not already containing an integral multiple of θ .
- **Safe-triangle invariant (successful):** Call a triangle safe if none of its angles is a positive integral multiple of θ . If $180^\circ/\theta$ is not an integer, every cut of a safe triangle has a safe child. Starting from an equilateral safe triangle, Shan-Yu can therefore avoid defeat forever.
- **Sanity checks:** `code/sanity_checks.py` checked the safe-child invariant on 130242 rational-grid cut configurations and the integer interval lemma on 18744 rational-grid triples. These computations are not used as proof.

Current best

The complete characterization is

$$\theta = \frac{180^\circ}{n} \quad \text{for an integer } n \geq 2.$$

For these and only these values Mulan has a finite winning strategy. In fact, the proof gives a bound depending only on n for the number of cuts.

Full proof

We prove both directions. All angles below are measured in degrees.

1. The effect of a cut

Let a triangle ABC have angles A, B, C . Suppose that P is an interior point of BC , and put

$$x = \angle BAP.$$

Then $0 < x < A$. The angle triples of the two resulting triangles ABP and ACP are, respectively,

$$(B, x, 180^\circ - B - x) \quad (1)$$

and

$$(C, A - x, B + x). \quad (2)$$

Indeed, the first two entries displayed in each triple are the inherited angles at the original vertices, and the third follows from the angle sum of a triangle, using $A + B + C = 180^\circ$.

Conversely, every x with $0 < x < A$ can be realized by a legal cut: the ray from A inside $\angle BAC$ which makes angle x with AB meets the opposite side BC at an interior point. Thus, in using (1)–(2), it is enough to verify $0 < x < A$.

2. Triangles containing integral multiples of θ

We first establish a forcing observation.

Lemma 1. If the current triangle has an angle $m\theta$, where m is a positive integer, then Mulan can force a win after at most $m - 1$ further cuts.

Proof. We use induction on m . For $m = 1$, the stopping condition already holds, so no cut is needed.

Let $m \geq 2$. From the vertex with angle $m\theta$, Mulan makes the cut which divides that angle into

$$\theta + (m - 1)\theta.$$

Both parts are positive, so this is a legal cut by the observation above. One child has an angle θ , while the other has an angle $(m - 1)\theta$. If Shan-Yu retains the first, Mulan wins at the next stopping check. If he retains the second, the induction hypothesis gives a win after at most $m - 2$ additional cuts. In either case at most $m - 1$ cuts are required. $\square$

3. Sufficiency

Suppose

$$180^\circ = n\theta \quad (3)$$

for an integer $n \geq 2$. We show that Mulan can win from every initial triangle.

If the current triangle has an angle equal to a positive integral multiple of θ , Lemma 1 applies. It remains to handle a triangle none of whose angles is an integral multiple of θ .

Divide its three angles by θ , obtaining positive, nonintegral real numbers a, b, c such that

$$a + b + c = n. \quad (4)$$

We need the following elementary claim.

Claim. After possibly relabeling a, b, c , there is an integer k such that

$$b < k < a + b. \quad (5)$$

Moreover, $1 \leq k \leq n - 1$.

Proof of the claim. First suppose at least one of a, b, c is greater than 1. Name such a number a , name either of the remaining numbers b , and take $k = \lceil b \rceil$. Because b is not an integer,

$$0 < k - b < 1 < a,$$

which proves (5).

Otherwise all three numbers are less than 1: they are not equal to 1 because they are nonintegral. Hence (4) gives $n < 3$. Since $n \geq 2$ is an integer, $n = 2$. Name the numbers in any order. We have $b < 1$, while

$$a + b = 2 - c > 1.$$

Thus $k = 1$ satisfies (5).

Finally, in either case, (5) and positivity give $k > b > 0$, and

$$k < a + b = n - c < n.$$

Since k is an integer, $1 \leq k \leq n - 1$. This proves the claim. $\square$

Label the original triangle so that its angles are

$$A = a\theta, \quad B = b\theta, \quad C = c\theta,$$

where (5) holds. Mulan cuts from the vertex with angle A , choosing

$$x = (k - b)\theta.$$

The inequalities (5) give

$$0 < x < a\theta = A,$$

so this is a legal cut. By (1)–(2), one child has at the cut point the angle

$$B + x = b\theta + (k - b)\theta = k\theta, \quad (6)$$

while the other has at the cut point the angle

$$180^\circ - B - x = n\theta - k\theta = (n - k)\theta. \quad (7)$$

Both k and $n - k$ are positive integers. Whichever child Shan-Yu retains, it therefore contains a positive integral multiple of θ , so Lemma 1 gives Mulan a win after finitely many further cuts. This proves sufficiency.

4. Necessity

Now suppose that

$$\frac{180^\circ}{\theta} \notin \mathbb{Z}. \quad (8)$$

Call a triangle **safe** if none of its angles is a positive integral multiple of θ . We show that after every cut of a safe triangle, at least one child is safe.

Let the safe parent have angles A, B, C , and consider a cut from the vertex with angle A . With the notation of (1)–(2), its children have angle triples

$$(B, x, 180^\circ - B - x), \quad (C, A - x, B + x). \quad (9)$$

Assume for contradiction that both children are not safe. The inherited angles B and C are not integral multiples of θ , because the parent is safe. Hence there are positive integers p, q such that one equality from each of the following rows holds:

$$x = p\theta \quad \text{or} \quad 180^\circ - B - x = p\theta, \quad (10)$$

$$A - x = q\theta \quad \text{or} \quad B + x = q\theta. \quad (11)$$

There are four possible pairings:

1. If $x = p\theta$ and $A - x = q\theta$, then

$$A = (p + q)\theta,$$

contradicting safety of the parent.

2. If $x = p\theta$ and $B + x = q\theta$, then

$$B = (q - p)\theta.$$

Since $B > 0$, $q - p$ is a positive integer, again contradicting safety.

3. If $180^\circ - B - x = p\theta$ and $A - x = q\theta$, then, using $A + B + C = 180^\circ$,

$$C = (180^\circ - B - x) - (A - x) = (p - q)\theta.$$

Since $C > 0$, this contradicts safety.

4. If $180^\circ - B - x = p\theta$ and $B + x = q\theta$, then adding gives

$$180^\circ = (p + q)\theta,$$

contradicting (8).

All possibilities are impossible. Therefore at least one child of every cut of a safe triangle is safe.

Finally, Shan-Yu can choose an equilateral initial triangle. It is safe: if its angle 60° were $m\theta$ for some positive integer m , then

$$\frac{180^\circ}{\theta} = 3m$$

would be an integer, contrary to (8). After every cut, Shan-Yu retains a safe child, which exists by the preceding argument. Every retained triangle is nondegenerate because the cut point is not a vertex. Since a safe triangle has no angle θ , the stopping condition never occurs. Thus Mulan cannot guarantee a finite victory when (8) holds.

Combining sufficiency and necessity, and observing that $0^\circ < \theta < 180^\circ$ makes an integral value $180^\circ/\theta$ at least 2, yields exactly

$$\boxed{\theta = 180^\circ/n \quad (n = 2, 3, 4, \dots)}.$$

IMO 2026 — Problem 5

Solution by GPT-5.6 Sol (xhigh reasoning)

2026-07-18

Source: results/gpt-5.6-sol-xhigh/problem-05/current.md — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Approaches tried

- **Direct equality and orbit analysis (successful):** Substituting $x = f(y)$ makes the two outer expressions equal, hence forces $f(f(y)) = 2f(y) - y$. Forward iteration then shows that $f(y) - y$ is nonnegative and remains constant along the whole forward orbit.
- **Asymptotic comparison of two orbits (successful):** If two points have positive displacements $a = f(u) - u$ and $b = f(v) - v$, their forward orbits are arithmetic progressions. Choosing far-out terms of the two progressions that remain a bounded distance apart makes the RMS-AM gap tend to zero, forcing $a = b$.
- **Topological elimination of mixed behavior (successful):** Once every displacement is either 0 or one common $c > 0$, the first inequality shows that the zero-displacement and c -displacement sets are both open. Connectedness of $\mathbb{R}_{>0}$ rules out a mixture.
- **Ratio/asymptotic-bound exploration (abandoned):** Writing $f(t) = tr(t)$ gives pairwise quadratic constraints and rough bounds, but these do not by themselves expose the exact solutions as efficiently as the equality substitution $x = f(y)$.

Current best

All solutions are

$$\boxed{f(x) = x + c \quad \text{for every } x > 0, \quad c \geq 0.}$$

The proof below derives an arithmetic-progression structure for every forward orbit, proves that all positive orbit increments coincide, and then excludes coexistence of zero and positive increments.

Full proof

Put

$$d(t) = f(t) - t \quad (t > 0).$$

We first determine the forward orbits of f . In the given inequalities, fix $y > 0$ and set $x = f(y)$. Both outer expressions then equal $f(y)$:

$$\sqrt{\frac{f(y)^2 + f(y)^2}{2}} = f(y) \quad \text{and} \quad \sqrt{f(y)f(y)} = f(y).$$

Consequently the middle expression must also equal $f(y)$, so

$$\frac{f(f(y)) + y}{2} = f(y), \quad \text{or equivalently} \quad f(f(y)) = 2f(y) - y. \quad (1)$$

For any fixed $t > 0$, define $t_0 = t$ and $t_{n+1} = f(t_n)$ for $n \geq 0$. Applying (1) to t_n gives

$$t_{n+2} = 2t_{n+1} - t_n.$$

Thus, by induction,

$$t_n = t + n(f(t) - t) = t + nd(t) \quad (n \geq 0). \quad (2)$$

Every t_n belongs to $\mathbb{R}_{>0}$. If $d(t) < 0$, the right-hand side of (2) is nonpositive for all sufficiently large integers n , a contradiction. Hence

$$d(t) \geq 0 \quad \text{for every } t > 0. \quad (3)$$

Moreover, (2) shows that along this orbit

$$f(t + nd(t)) = t + (n + 1)d(t), \quad (4)$$

so its displacement remains equal to $d(t)$.

We next prove that any two positive displacements are equal. Let $u, v > 0$, and write

$$a = d(u) > 0, \quad b = d(v) > 0.$$

Suppose, without loss of generality, that $a > b$. By (4), for all nonnegative integers n, m ,

$$f(u + na) = u + (n + 1)a, \quad f(v + mb) = v + (m + 1)b. \quad (5)$$

Choose integers $n \rightarrow \infty$ and

$$m_n = \left\lfloor \frac{u + na - v - b}{b} \right\rfloor.$$

For all sufficiently large n , $m_n \geq 0$. Set

$$p_n = u + na, \quad q_n = v + (m_n + 1)b.$$

The definition of m_n gives

$$0 \leq p_n - q_n < b, \quad (6)$$

and both p_n and q_n tend to infinity.

Apply the first given inequality with $x = p_n$ and $y = v + m_n b$. By (5), $f(x) = p_n + a$ and $f(y) = q_n$, and hence

$$p_n + q_n + (a - b) = f(x) + y \leq \sqrt{2(p_n^2 + q_n^2)}.$$

It follows that

$$0 < a - b \leq \sqrt{2(p_n^2 + q_n^2)} - p_n - q_n = \frac{(p_n - q_n)^2}{\sqrt{2(p_n^2 + q_n^2)} + p_n + q_n}. \quad (7)$$

By (6), the numerator on the right is less than b^2 , whereas its denominator tends to infinity. The right-hand side of (7) therefore tends to 0, contradicting $a - b > 0$. Interchanging u and v similarly rules out $b > a$. Thus

$$d(u) = d(v) \quad \text{whenever } d(u) > 0 \text{ and } d(v) > 0. \quad (8)$$

If d is identically zero, then $f(t) = t$, which is one of the claimed functions. Otherwise, by (8), there is a constant $c > 0$ such that

$$d(t) \in \{0, c\} \quad \text{for every } t > 0. \quad (9)$$

Define

$$A = \{t > 0 : d(t) = 0\}, \quad B = \{t > 0 : d(t) = c\}.$$

We show that both A and B are open in $\mathbb{R}_{>0}$. If this were false at some point of either set, there would be sequences $r_k \in B$ and $s_k \in A$ with $r_k - s_k \rightarrow 0$ and with both sequences converging to the same positive limit L (one can keep whichever of the two points is fixed and choose points of the other set approaching it). Apply the first inequality with $x = r_k$ and $y = s_k$. Since $f(r_k) = r_k + c$ and $f(s_k) = s_k$, it yields

$$r_k + s_k + c = f(r_k) + s_k \leq \sqrt{2(r_k^2 + s_k^2)}. \quad (10)$$

Taking $k \rightarrow \infty$ in (10) gives $2L + c \leq 2L$, impossible. Therefore every point of either A or B has a neighborhood containing no point of the other set; both sets are open.

The disjoint open sets A and B have union $\mathbb{R}_{>0}$. Since $\mathbb{R}_{>0}$ is connected and B is nonempty, we must have $A = \emptyset$. Hence $d(t) = c$ for every $t > 0$. Together with the identically-zero case, this proves that every possible function has the form

$$f(t) = t + c$$

for some constant $c \geq 0$.

Finally, every such function does satisfy the inequalities. Indeed, if $f(t) = t + c$ with $c \geq 0$, then for all $x, y > 0$,

$$\sqrt{\frac{x^2 + f(y)^2}{2}} = \sqrt{\frac{x^2 + (y + c)^2}{2}} \geq \frac{x + (y + c)}{2} = \frac{f(x) + y}{2} \geq \sqrt{x(y + c)} = \sqrt{xf(y)},$$

where the two inequalities are respectively the root-mean-square–arithmetic-mean and arithmetic-mean–geometric-mean inequalities for the positive numbers x and $y + c$. This completes the characterization.

IMO 2026 — Problem 6

Solution by GPT-5.6 Sol (xhigh reasoning)

2026-07-18

Source: `results/gpt-5.6-sol-xhigh/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Solution document

Status

solved

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Approaches tried

- Reformulated the sequence as the result of scanning all integers above a_1 and retaining exactly those non-coprime to every previously retained integer.
- A direct attempt to bound all prime divisors of the terms fails: terms can have arbitrarily large irrelevant prime factors.
- The successful route is instead to consider the inclusion-minimal prime supports of retained integers. They form a self-blocking intersecting family. An arithmetic descent shows that every prime occurring in a minimal support also occurs in a minimal support whose remaining radical is at most a_1 . There are only finitely many such small cores, and a disjoint witness

shows that each core can be paired with only finitely many primes. Hence the set of relevant primes is finite.

Current best

Let $\mathcal{H}$ be the family of inclusion-minimal prime supports of sequence terms. The proof establishes that $U = \bigcup \mathcal{H}$ is finite. Consequently the sequence is exactly the increasing enumeration, beginning at a_1 , of integers divisible by $\prod_{p \in H} p$ for at least one $H \in \mathcal{H}$. This condition is periodic modulo $L = \prod_{p \in U} p$. If T is the number of admissible residue classes modulo L , then $a_{n+T} = a_n + L$ for every n .

Full proof

Put $m = a_1$, and for every integer $x > 1$ let $P(x)$ denote the finite set of prime divisors of x .

1. The self-blocking family of prime supports

The terms a_n are strictly increasing, so the sequence is unbounded. We first record the following consequence of the defining rule:

Scanning property. If an integer $x > m$ is not a term of the sequence, then there is a term $a_i < x$ such that $\gcd(x, a_i) = 1$.

Indeed, choose n such that

$$a_n < x < a_{n+1}.$$

If x had a common prime divisor with each of $a_1, \dots, a_n$, it would be an eligible choice after a_n , contradicting the fact that a_{n+1} is the smallest eligible integer. This proves the scanning property.

Let

$$\mathcal{G} = \{P(a_n) : n \geq 1\}$$

and define its *blocker* (among finite sets of primes) by

$$\mathcal{F} = \{S : S \text{ is a finite set of primes and } S \cap G \neq \emptyset \text{ for every } G \in \mathcal{G}\}.$$

Any two terms of the sequence have gcd greater than 1, so any two members of $\mathcal{G}$ intersect. It follows that $\mathcal{G} \subseteq \mathcal{F}$.

In fact,

$$\boxed{\mathcal{G} = \mathcal{F}.} \quad (1)$$

To prove the reverse inclusion, take $S \in \mathcal{F}$. The set S is nonempty, because it must intersect $P(a_1)$. Choose an integer $x > m$ with $P(x) = S$; for example, a sufficiently large power of $\prod_{p \in S} p$ has these properties. Since S intersects every member of $\mathcal{G}$, the integer x has gcd greater than 1 with every term of the sequence. As the sequence is unbounded, we may choose n such that

$$a_n < x \leq a_{n+1}.$$

Then x is eligible after a_n , so the minimality in the recurrence gives $a_{n+1} \leq x$. Hence $x = a_{n+1}$, and therefore $S = P(x) \in \mathcal{G}$. This proves (1).

Thus $\mathcal{F}$ has the following properties:

1. it is **upward-closed**: if $S \in \mathcal{F}$ and $S \subseteq S'$, where S' is finite, then $S' \in \mathcal{F}$;
2. it is **intersecting**: any two of its members intersect;
3. it is **self-blocking**:

$$S \in \mathcal{F} \iff S \cap F \neq \emptyset \text{ for every } F \in \mathcal{F}. \quad (2)$$

Here (2) follows from the definition of $\mathcal{F}$ and (1). In particular, whenever a finite set S does not belong to $\mathcal{F}$, there exists $F \in \mathcal{F}$ disjoint from S .

Let $\mathcal{H}$ be the family of inclusion-minimal members of $\mathcal{F}$. Every member F of $\mathcal{F}$ contains a member of $\mathcal{H}$: among the finitely many subsets of F which belong to $\mathcal{F}$, choose an inclusion-minimal one.

2. Finiteness of the relevant primes

Call the primes in

$$U = \bigcup_{H \in \mathcal{H}} H$$

the **relevant primes**. We prove that U is finite.

Fix $p \in U$, and choose $H \in \mathcal{H}$ containing p . Set

$$c(H, p) = \prod_{q \in H \setminus \{p\}} q,$$

where an empty product is 1. We first prove that, after possibly replacing H by another member of $\mathcal{H}$ which still contains p , we may arrange that

$$c(H, p) \leq m. \quad (3)$$

Suppose that the present H does not satisfy (3), and put

$$D = H \setminus \{p\}, \quad x = c(H, p) > m.$$

The minimality of H gives $D \notin \mathcal{F}$. By (1), no term of the sequence has prime support D ; in particular, x is not a term. By the scanning property there is a term $y = a_j < x$ with

$$P(y) \cap D = \emptyset.$$

The set $P(y)$ belongs to $\mathcal{F}$, so it contains some $K \in \mathcal{H}$. Hence $K \cap D = \emptyset$. On the other hand, $K, H \in \mathcal{F}$, and $\mathcal{F}$ is intersecting. Since $H = D \cup \{p\}$, these two facts force $p \in K$. Moreover, the squarefree number $p c(K, p)$ divides y , and therefore

$$c(K, p) \leq \frac{y}{p} < x = c(H, p). \quad (4)$$

Thus replacing H by K strictly decreases the positive integer $c(H, p)$ while preserving both $H \in \mathcal{H}$ and $p \in H$. Repeating this operation must terminate, and it can terminate only with (3). This proves the claim.

There are only finitely many finite sets C of primes satisfying

$$\prod_{q \in C} q \leq m, \quad (5)$$

because every prime in such a set is at most m . By (3), every $p \in U$ admits at least one such set C for which

$$C \cup \{p\} \in \mathcal{H}, \quad p \notin C. \quad (6)$$

We show that each fixed C satisfying (5) can occur in (6) for only finitely many primes p .

If (6) holds, the inclusion-minimality of $C \cup \{p\}$ implies that $C \notin \mathcal{F}$. By (2), we may therefore fix some $F_C \in \mathcal{F}$ with

$$F_C \cap C = \emptyset.$$

For every prime p satisfying (6), the two members F_C and $C \cup \{p\}$ of the intersecting family $\mathcal{F}$ must intersect. Since F_C is disjoint from C , this forces $p \in F_C$. The set F_C is finite, so only finitely many such p exist.

There are finitely many possible C , and each yields only finitely many possible p . Hence U is finite. Since every $H \in \mathcal{H}$ is a subset of the finite set U , the family $\mathcal{H}$ itself is finite as well.

3. Periodicity

For every finite set S of primes, upward-closure and the definition of $\mathcal{H}$ give

$$S \in \mathcal{F} \iff H \subseteq S \text{ for some } H \in \mathcal{H}. \quad (7)$$

We now characterize the terms of the sequence. For every integer $x \geq m$,

$$x \text{ is a term of the sequence} \iff H \subseteq P(x) \text{ for some } H \in \mathcal{H}. \quad (8)$$

The forward implication follows from $P(x) \in \mathcal{G} = \mathcal{F}$ and (7). Conversely, suppose the right-hand side holds. Then $P(x) \in \mathcal{F}$, so x has gcd greater than 1 with every term of the sequence. If $x = m$, it is already the first term. If $x > m$ and were not a term, the scanning property would supply an earlier term coprime to x , a contradiction. This proves (8).

The set U is nonempty because $\mathcal{F}$ contains $P(m)$ and hence $\mathcal{H}$ is nonempty. Define

$$L = \prod_{p \in U} p.$$

For any positive integer x and any $H \in \mathcal{H}$, all primes of H divide L , so

$$H \subseteq P(x) \iff H \subseteq P(x + L).$$

Thus the condition on the right of (8) is periodic modulo L .

Let T be the number of integers $r \in \{1, 2, \dots, L\}$ satisfying that condition. This is positive: the integer L itself satisfies it because it is divisible by every prime in every $H \in \mathcal{H}$. By periodicity, every interval of L consecutive integers contains exactly T integers satisfying the condition.

Fix $n \geq 1$. By (8), among the integers

$$a_n + 1, a_n + 2, \dots, a_n + L$$

there are exactly T terms of the sequence. Also $a_n + L$ is one of them, since a_n is a term and the condition in (8) is periodic modulo L . Therefore $a_n + L$ is precisely the T -th term after a_n , that is,

$$\boxed{a_{n+T} = a_n + L}$$

for every positive integer n . Both T and L are positive, completing the proof.